REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 11 Advanced | Module 8 Lesson 3 | Equilibrium

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Static equilibrium: ΣF = 0 and Στ = 0. xCM = Σmixi/Σmi. Choose a convenient pivot and state torque signs.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 18–24.

P01. Static equilibrium requires:

A. Zero net force only B. Zero net torque only
C. Zero mass D. Zero net force and zero net torque

Answer

D. Zero net force and zero net torque.

P02. An object's centre of mass is:

A. The point representing its overall mass distribution B. Always outside the object
C. Always at its highest point D. Always its geometric centre regardless of density

Answer

A. The point representing its overall mass distribution.

P03. A wider support base usually makes an object:

A. Less stable in every situation B. More stable against tipping
C. Weightless D. Have zero moment of inertia

Answer

B. More stable against tipping.

P04. An object starts to tip when the vertical line through its centre of mass:

A. Passes through the centre of the support base B. Becomes horizontal
C. Passes outside the support base D. Always meets the geometric centre

Answer

C. Passes outside the support base.

P05. The Coriolis force is described in a rotating reference frame as:

A. A new fundamental interaction B. The object's weight only
C. A force that must exist in every inertial frame D. An apparent deflection-related force

Answer

D. An apparent deflection-related force.

P06. A balanced lever has downward force 40 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

A. 40 N B. 10 N
C. 20 N D. -20 N

Answer

C. 20 N. Equal opposing torques: F1r1 = F2r2. F2 = 40(0.30)/0.60 = 20 N.

P07. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 4 m lie on a light rod. Find xCM.

A. 4.8 m B. 1.2 m
C. -2.4 m D. 2.4 m

Answer

D. 2.4 m. $x_{CM} = \Sigma mx/\Sigma m = (2.0 \times 0 + 3.0 \times 4)/(2.0 + 3.0) = 2.4$ m.

P08. A uniform horizontal beam of weight 80 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

A. 40 N B. 80 N
C. 20 N D. -40 N

Answer

A. 40 N. Symmetry gives equal reactions. R1 + R2 = W, so each R = 80/2 = 40 N.

P09. A light 4.0 m beam is supported at both ends. A 40 N downward load is 1.0 m from the left end. Find the right-support reaction.

A. 20 N B. 10 N
C. 5 N D. -10 N

Answer

B. 10 N. Take moments about left end: $R_{right} \times 4.0 = 40 \times 1.0$; $R_{right} = 10$ N.

P10. An object in translational equilibrium has weight 60 N downward and one upward force 20 N. Find the other upward force.

A. 80 N B. 20 N
C. 40 N D. -40 N

Answer

C. 40 N. $\Sigma F_y = 0$, so $F + 20 - 60 = 0$; $F = 40$ N.

P11. Distinguish translational from rotational equilibrium.

Answer

Translational equilibrium means $\Sigma F = 0$; rotational equilibrium means $\Sigma \tau = 0$ about the relevant axis. Static equilibrium requires both.

P12. How can the centre of mass of a flat irregular object be located experimentally?

Answer

Suspend it from different points and draw a vertical plumb line from each suspension point; the lines intersect at its centre of mass.

P13. Why can lowering centre of mass improve stability?

Answer

A larger tilt is generally needed before the centre-of-mass vertical passes beyond the support boundary.

P14. What is meant by a rotating reference frame?

Answer

A frame whose axes rotate relative to an inertial frame; apparent effects such as centrifugal and Coriolis forces can be used when analysing motion within it.

P15. Can a centre of mass lie outside the material of an object?

Answer

Yes. For example, the centre of mass of a uniform ring lies at its empty centre.

P16. A balanced lever has downward force 60 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

Answer

Equal opposing torques: $F_1r_1 = F_2r_2$. $F_2 = 60(0.30)/0.60 = 30$ N.

P17. A balanced lever has downward force 80 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

Answer

Equal opposing torques: $F_1r_1 = F_2r_2$. $F_2 = 80(0.30)/0.60 = 40$ N.

P18. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 6 m lie on a light rod. Find xCM.

Answer

$x_{CM} = \Sigma mx/\Sigma m = (2.0 \times 0 + 3.0 \times 6)/(2.0 + 3.0) = 3.6$ m.

P19. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 8 m lie on a light rod. Find xCM.

Answer

$x_{CM} = \Sigma mx/\Sigma m = (2.0 \times 0 + 3.0 \times 8)/(2.0 + 3.0) = 4.8$ m.

P20. A uniform horizontal beam of weight 120 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

Answer

Symmetry gives equal reactions. $R_1 + R_2 = W$, so each $R = 120/2 = 60$ N.

P21. A uniform horizontal beam of weight 160 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

Answer

Symmetry gives equal reactions. $R_1 + R_2 = W$, so each $R = 160/2 = 80$ N.

P22. A light 4.0 m beam is supported at both ends. A 60 N downward load is 1.0 m from the left end. Find the right-support reaction.

Answer

Take moments about left end: $R_{right} \times 4.0 = 60 \times 1.0$; $R_{right} = 15$ N.

P23. A light 4.0 m beam is supported at both ends. A 80 N downward load is 1.0 m from the left end. Find the right-support reaction.

Answer

Take moments about left end: $R_{right} \times 4.0 = 80 \times 1.0$; $R_{right} = 20$ N.

P24. An object in translational equilibrium has weight 90 N downward and one upward force 30 N. Find the other upward force.

Answer

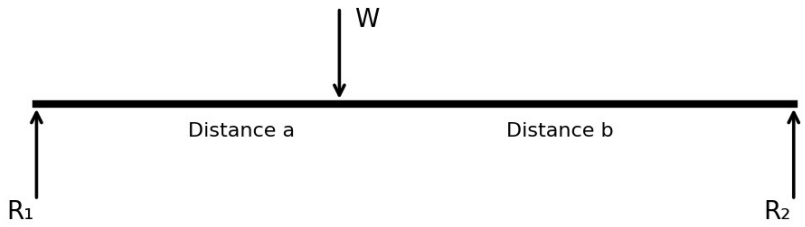
ΣFy = 0, so F + 30 − 90 = 0; F = 60 N.

P25. An object in translational equilibrium has weight 120 N downward and one upward force 40 N. Find the other upward force.

Answer

ΣFy = 0, so F + 40 − 120 = 0; F = 80 N.

P26. Draw a supported beam with one downward load. Explain how to find its two support reactions.



Answer

Label both upward reactions, the load and distances. Use ΣFy = 0 and take moments about one support to find the other reaction, then use force balance for the remaining one.

P27. A body has zero net force but a pair of equal opposite forces at different points. Must it be in equilibrium?

Answer

No. The forces can form a couple with a nonzero net torque. Both net force and net torque must vanish for static equilibrium.

P28. Compare two vehicles of equal width with different centre-of-mass heights. Which is more stable on a side slope?

Answer

The one with lower centre of mass is generally more stable because a larger tilt is needed for the vertical weight line to leave the wheel support base.

P29. Explain an observed sideways deflection of a rolling ball from the viewpoint of a rotating platform.

Answer

In the rotating frame the path appears deflected; the Coriolis effect accounts for this apparent force. In an inertial frame the motion is explained using the actual interactions.

P30. Why is taking moments about a support often a useful calculation choice?

Answer

The support reaction acts through that pivot and produces no torque about it, removing an unknown from the moment equation.